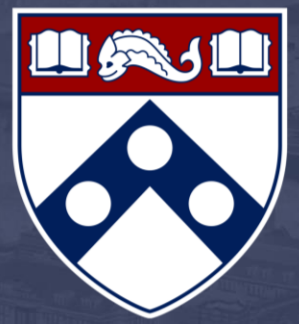
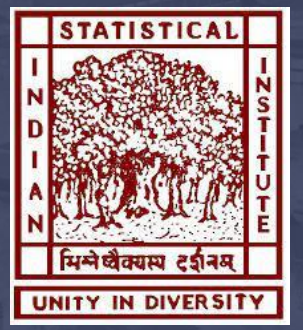


Predicting Survival Time of LGG Patients: Refining Feature Selection Methods

Mackenzie M., Neo K., Rosie D., and Sanchayan B.







UNIVERSITY OF MICHIGAN

Goals

- Compare different feature selection techniques to identify the one that yields the lowest Root Mean Squared Error (RMSE) in predicting survival time
- Develop a model that can be effectively generalized to newer data, mitigating the risk of overfitting and ensuring robust performance
- Address multicollinearity issues in the dataset by implementing appropriate methods to manage and control the impact of highly correlated predictor variables

Pipeline of Project



- Genomics
- Imaging

- LASSO
- Random Forests
- Boruta
- LM
- Bagged Trees

- Box-Cox

- Robustness

- Ridge Regression

- RMSE

Pipeline of Project



- Genomics
- Imaging

- LASSO
- Random Forests
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- LM
- Bagged Trees

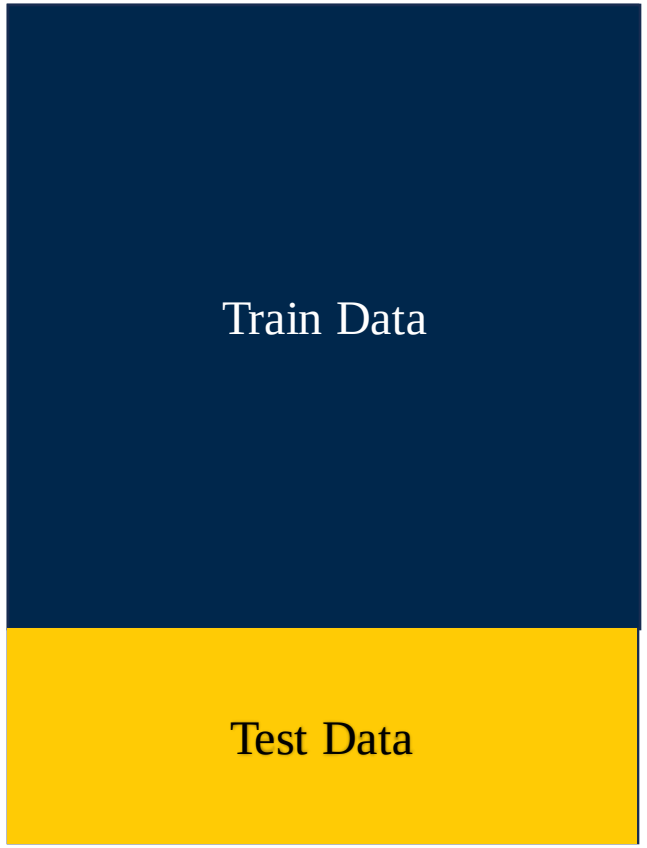
- Box-Cox

- Robustness

- Ridge Regression

- RMSE

Splitting Data



Pipeline of Project



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- Imaging

- LASSO
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- Boruta
- LM
- Bagged Trees

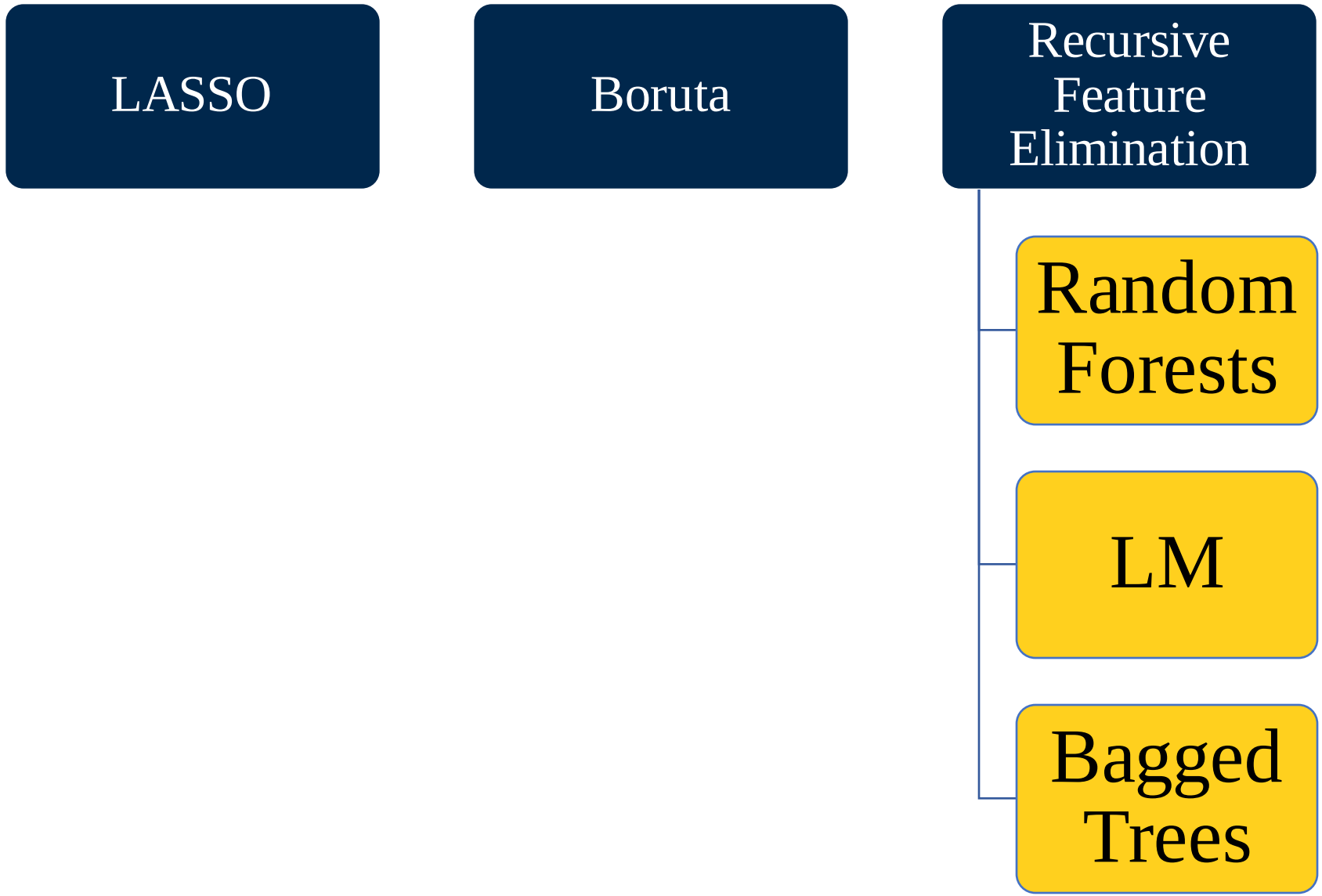
- Box-Cox

- Robustness

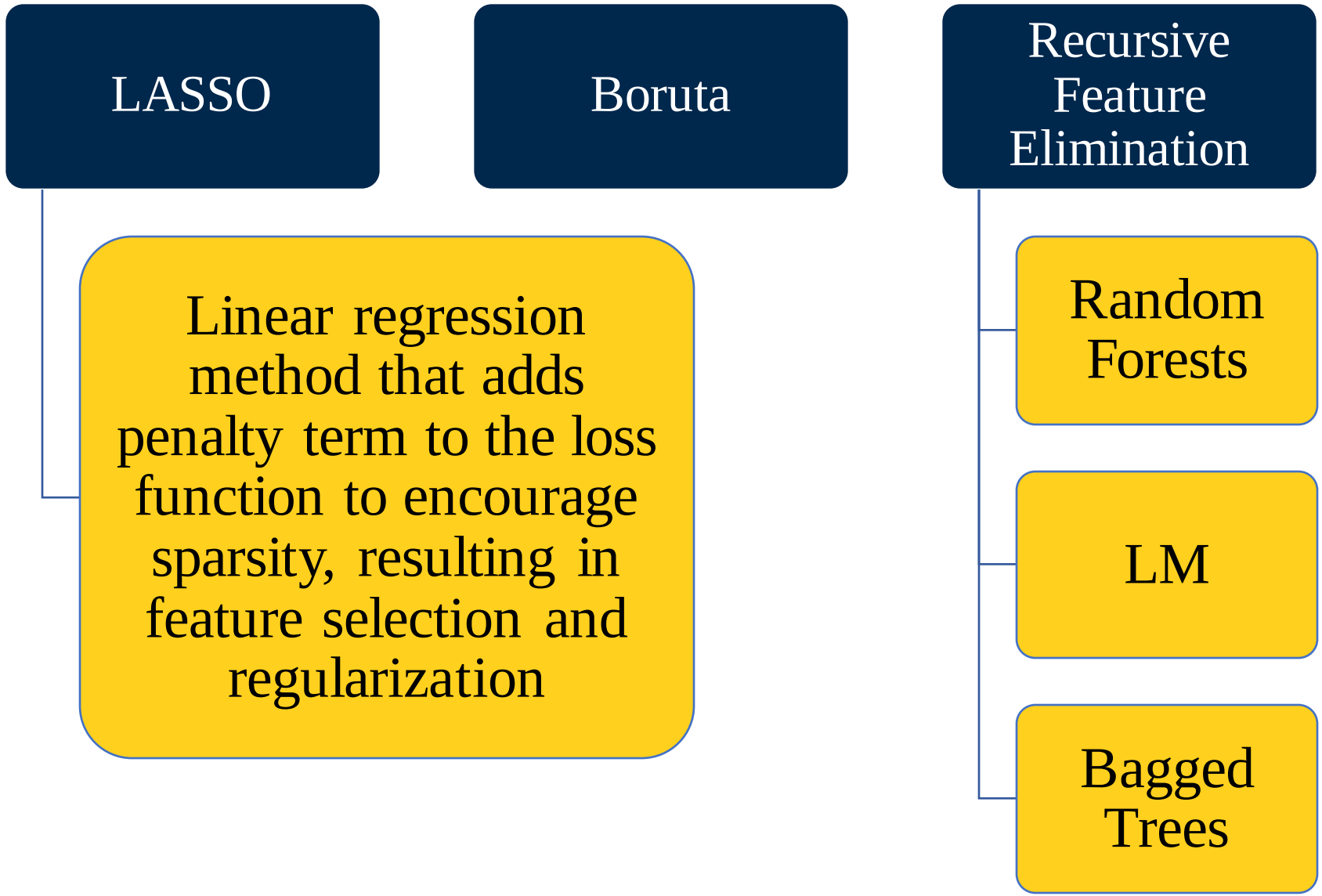
- Ridge Regression

- RMSE

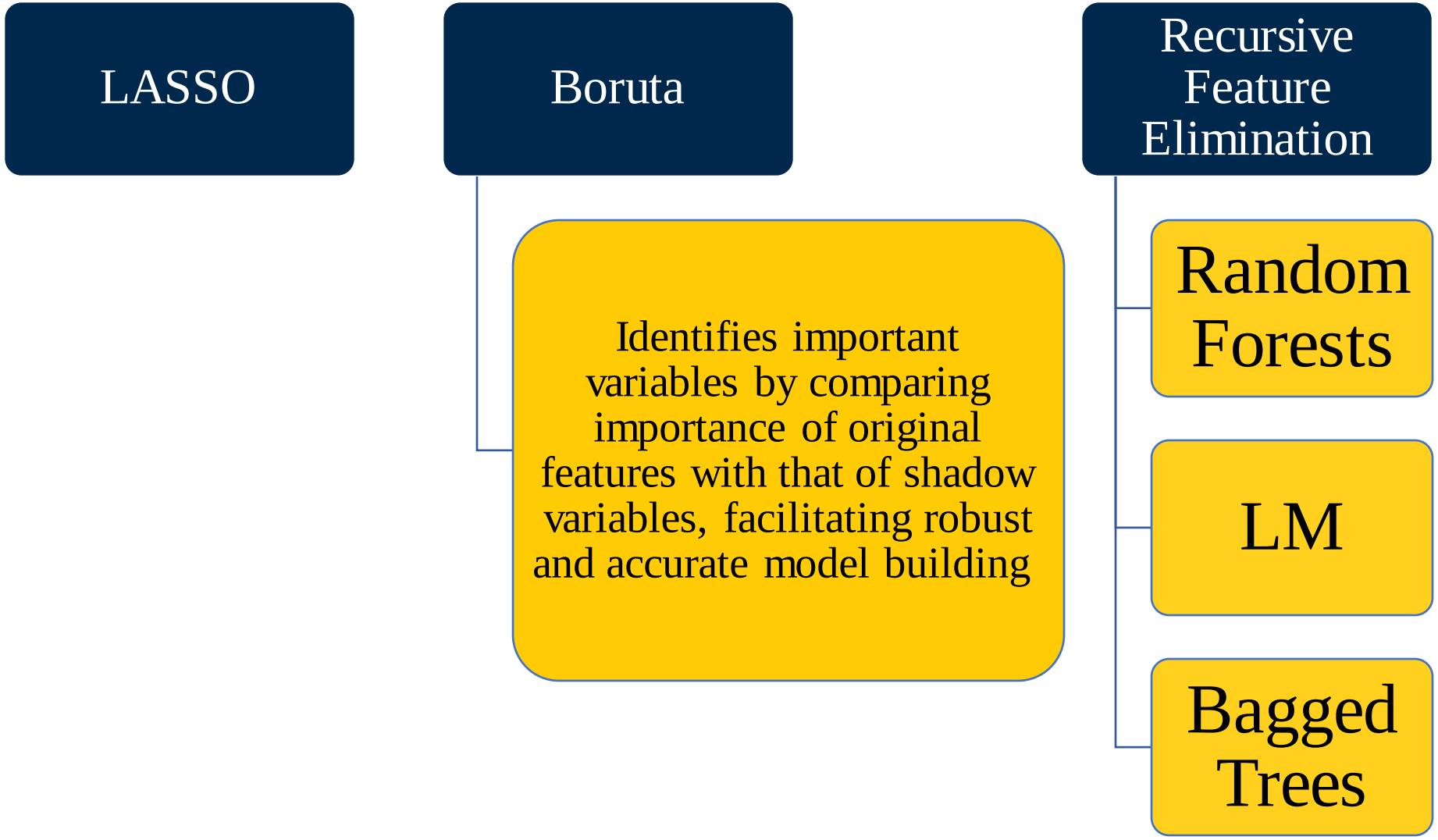
Feature Selection Methods



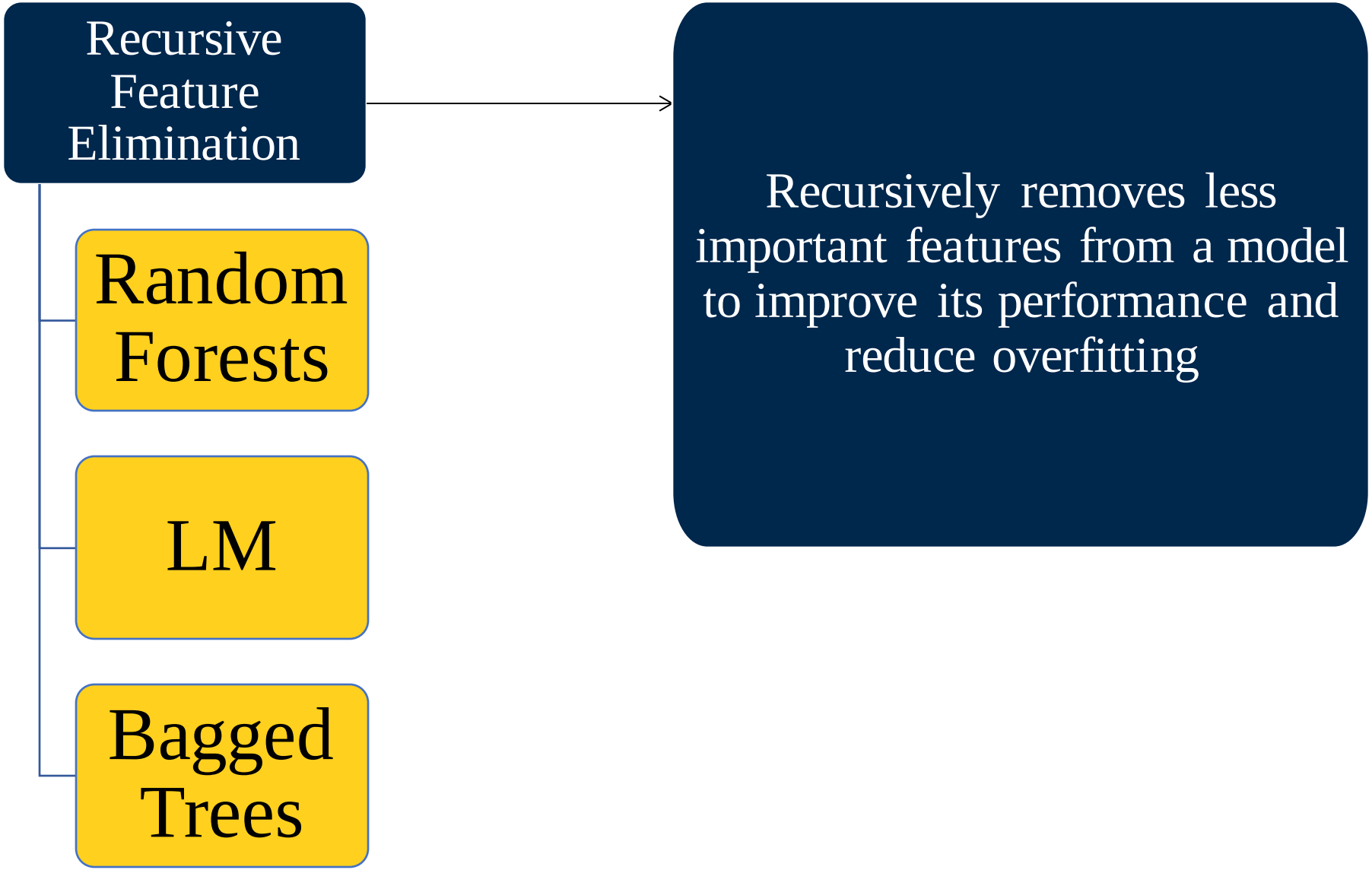
Feature Selection Methods



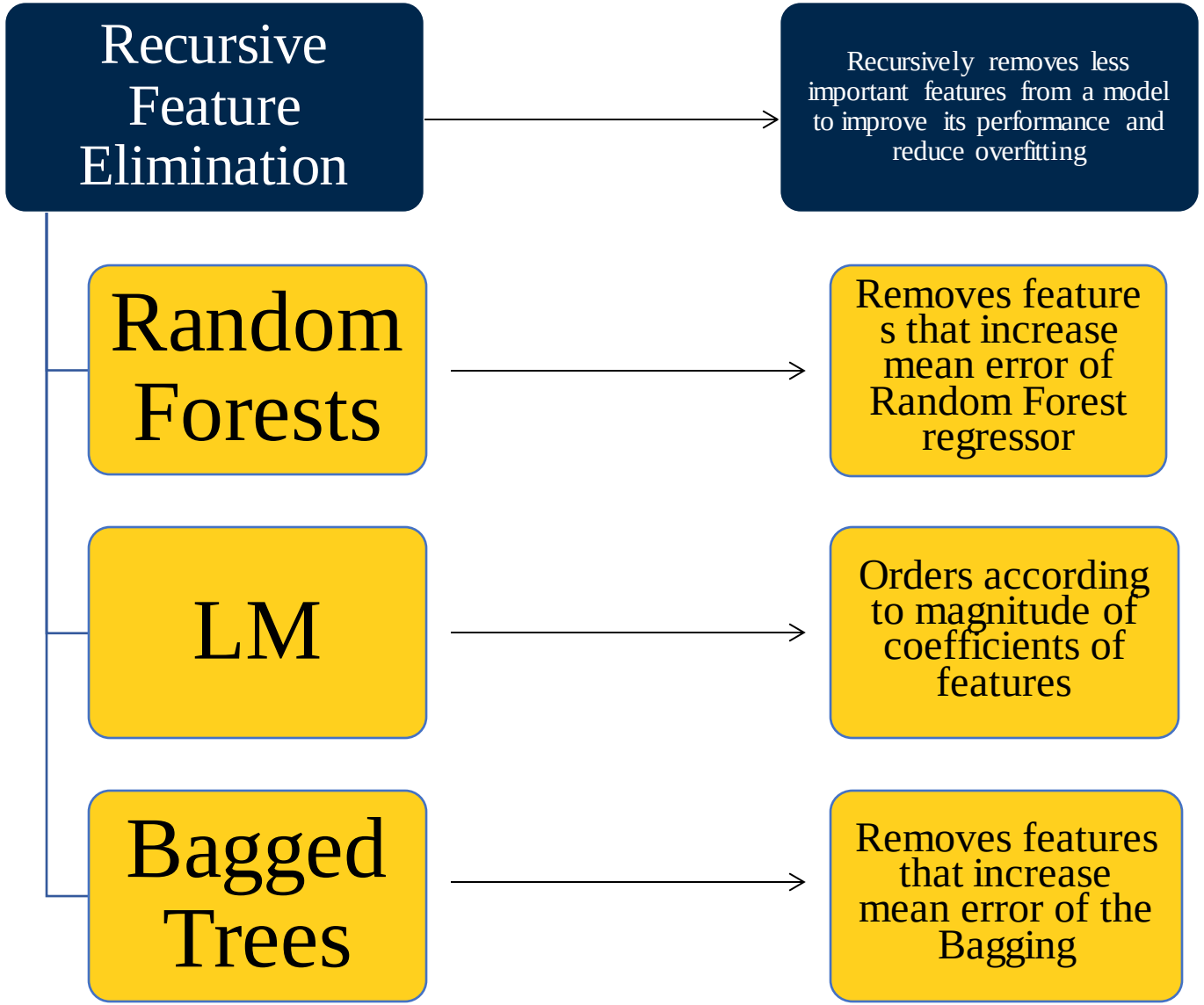
Feature Selection Methods



Feature Selection Methods



Feature Selection Methods



Pipeline of Project



- Genomics
- Imaging

- LASSO
- Random Forests
- Boruta
- LM
- Bagged Trees

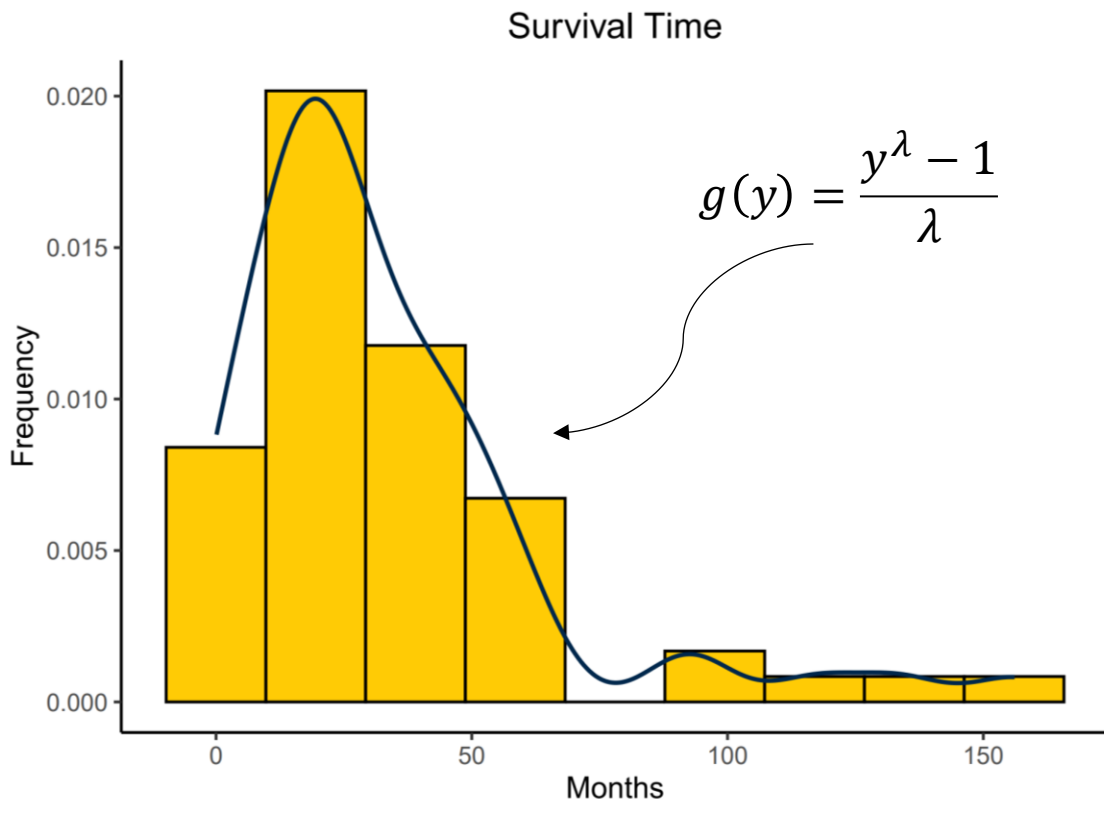
- Box-Cox

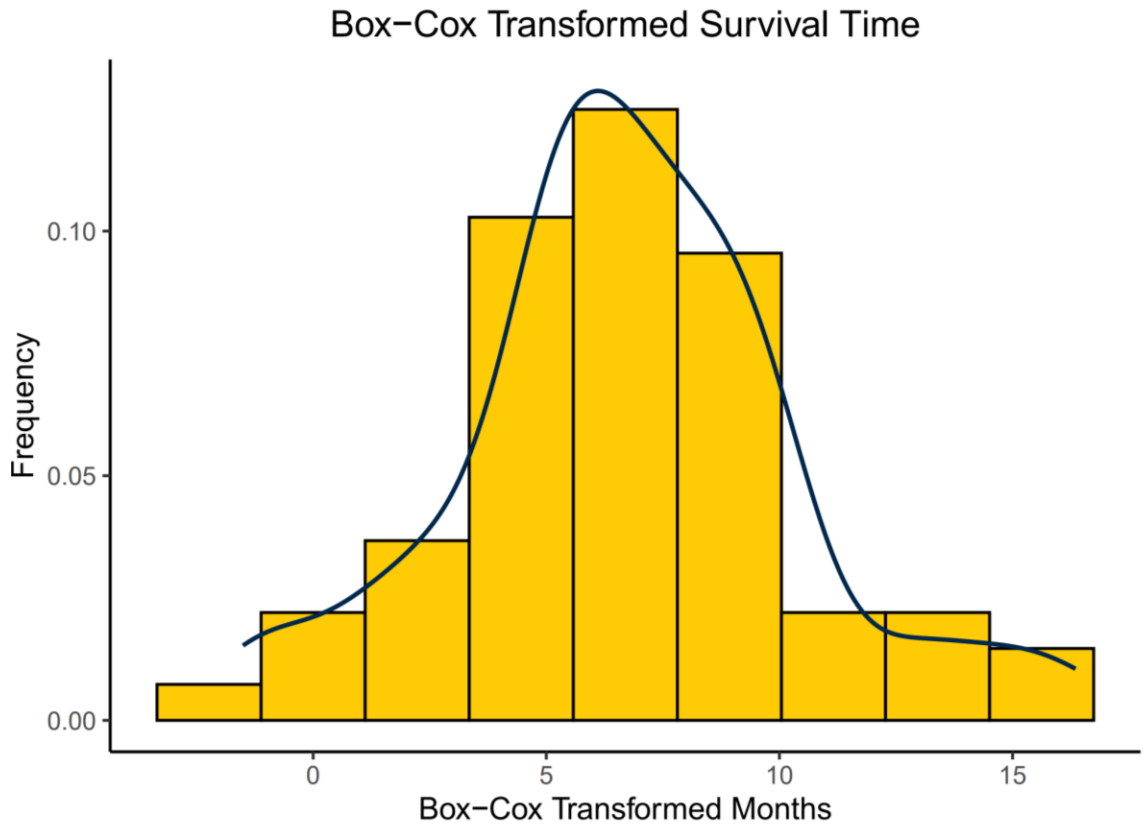
- Robustness

- Ridge Regression

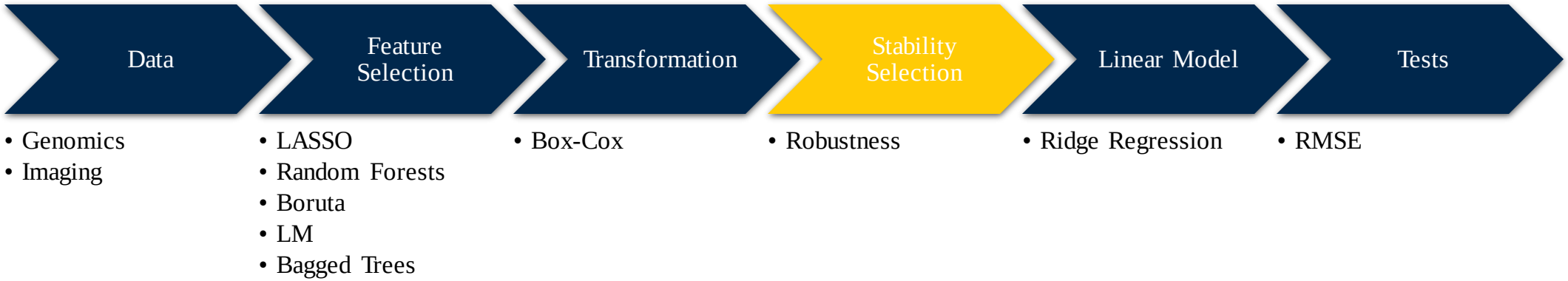
- RMSE

Survival Time Transformation





Pipeline of Project



Stability Selection

- Reduces overfitting and dependence on random seeds especially on high dimensional data

Steps:

- Run a feature selection method on a randomized subset of the training data
- Store the features provided by the given feature selection method
- Repeat n times with different randomized subsets S_i

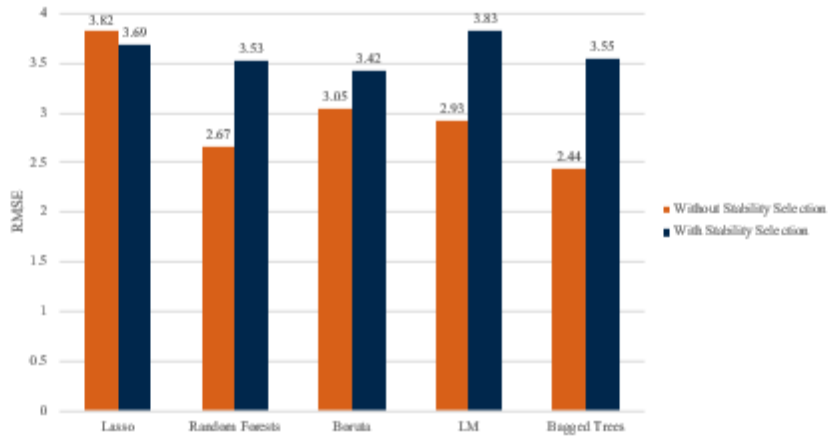
$$\widehat{\Pi_k} = P\{k \in \widehat{S}\} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\{k \in \widehat{S_i}\}$$

- Construct the stable set according to the following definition:

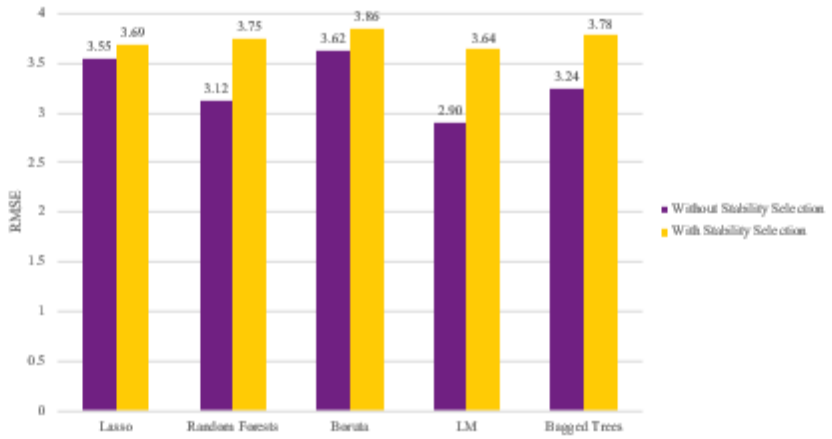
$$S^{stable} = \{k : \Pi_k > 0.8\}$$

Effect of Stability Selection

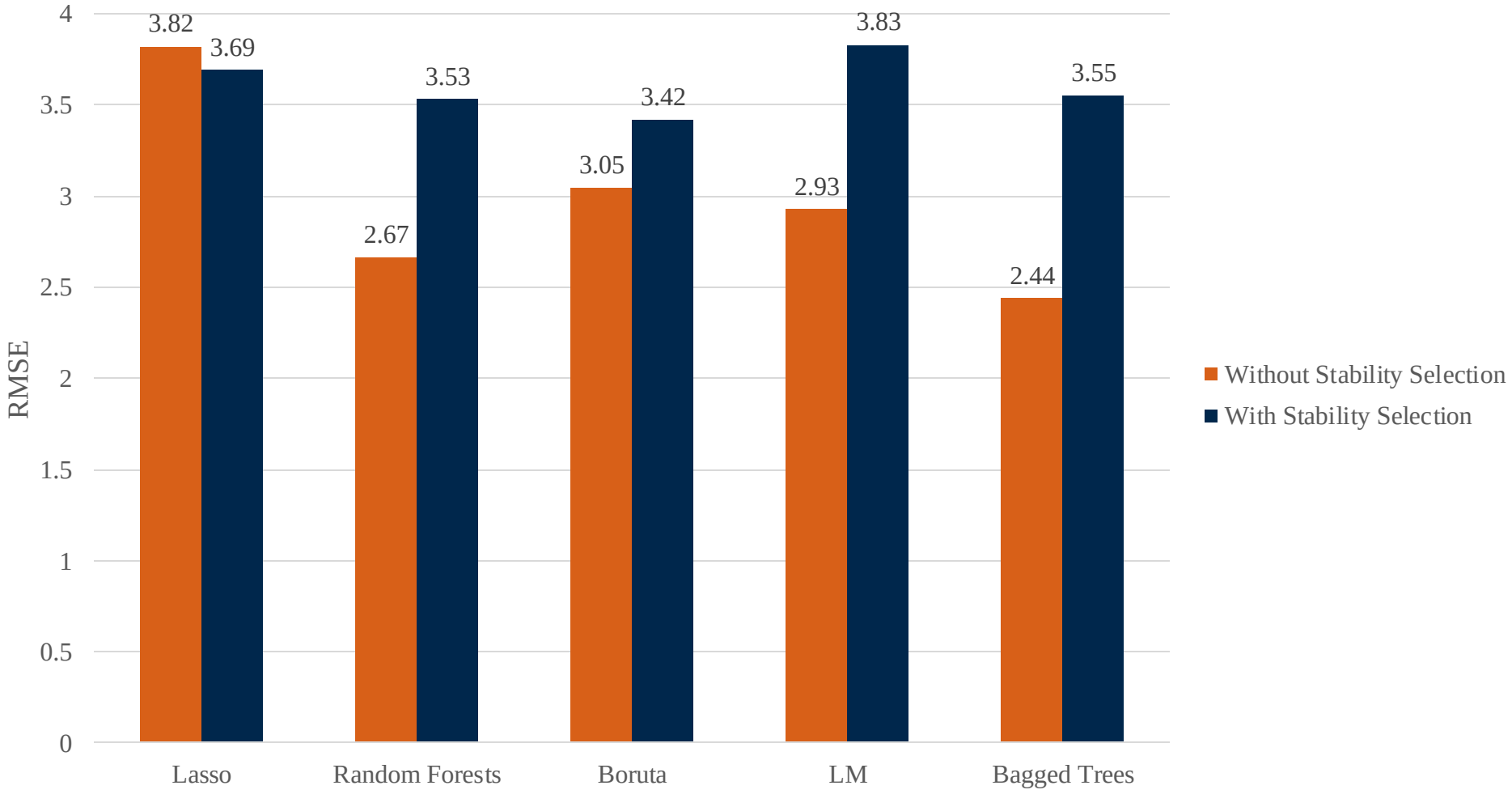
Gene Pathways



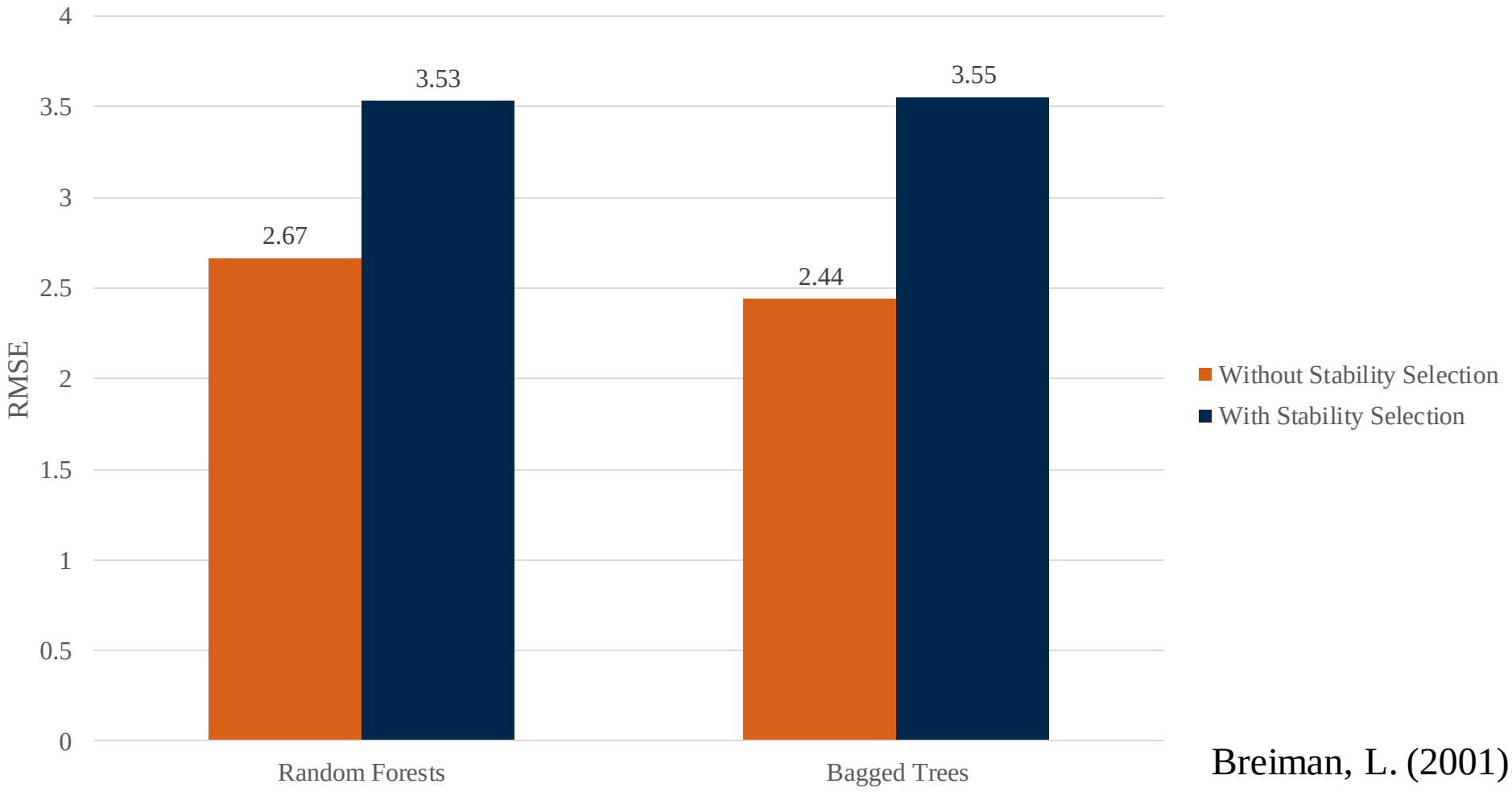
Imaging



Gene Pathways

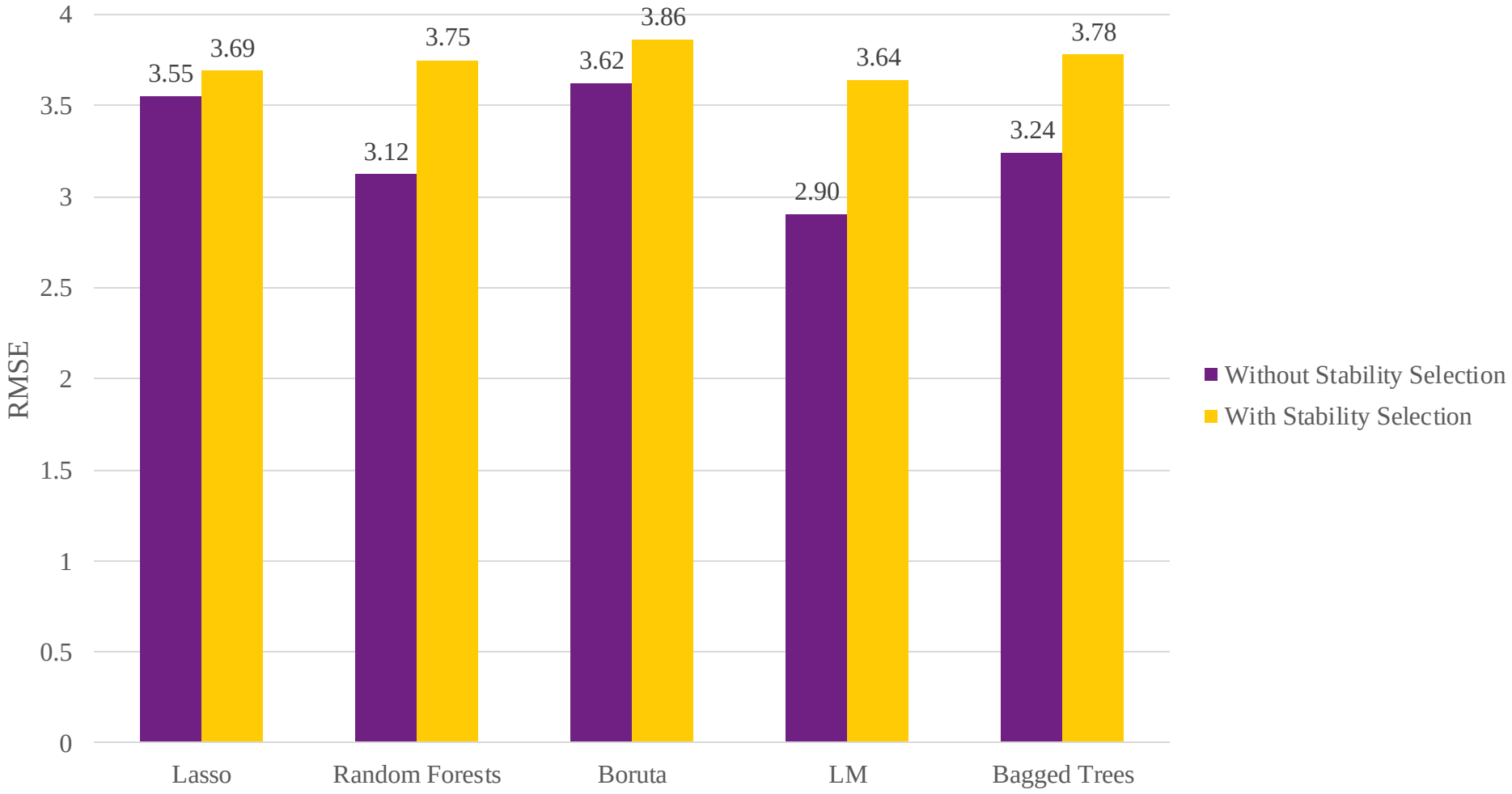


Gene Pathways

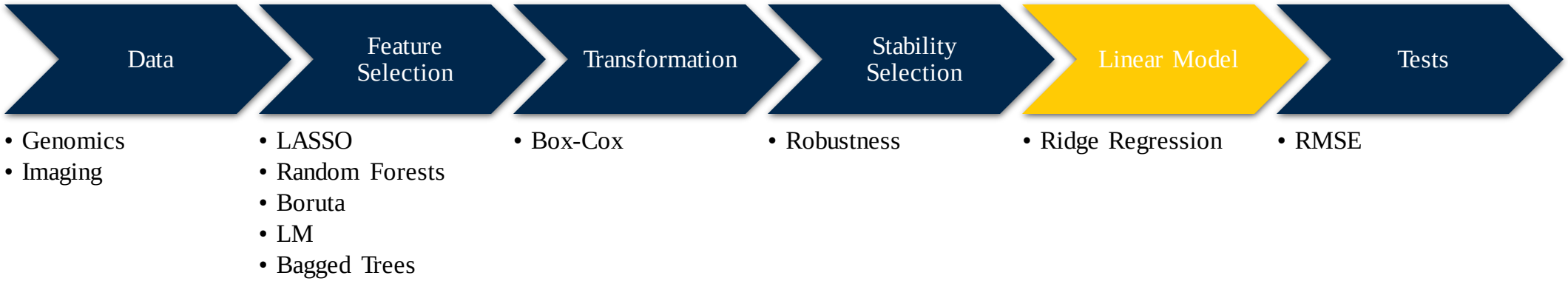


Breiman, L. (2001). Random forests. Machine learning, 45, 5-32.

Imaging


4
3.5
3
2.5
2
1.5
1
0.5
0
RMSE
3.55
3.69
3.12
3.75
3.62
3.86
2.90
3.64
3.24
3.78
Lasso
Random Forests
Boruta
LM
Bagged Trees
Without Stability Selection
With Stability Selection

Pipeline of Project

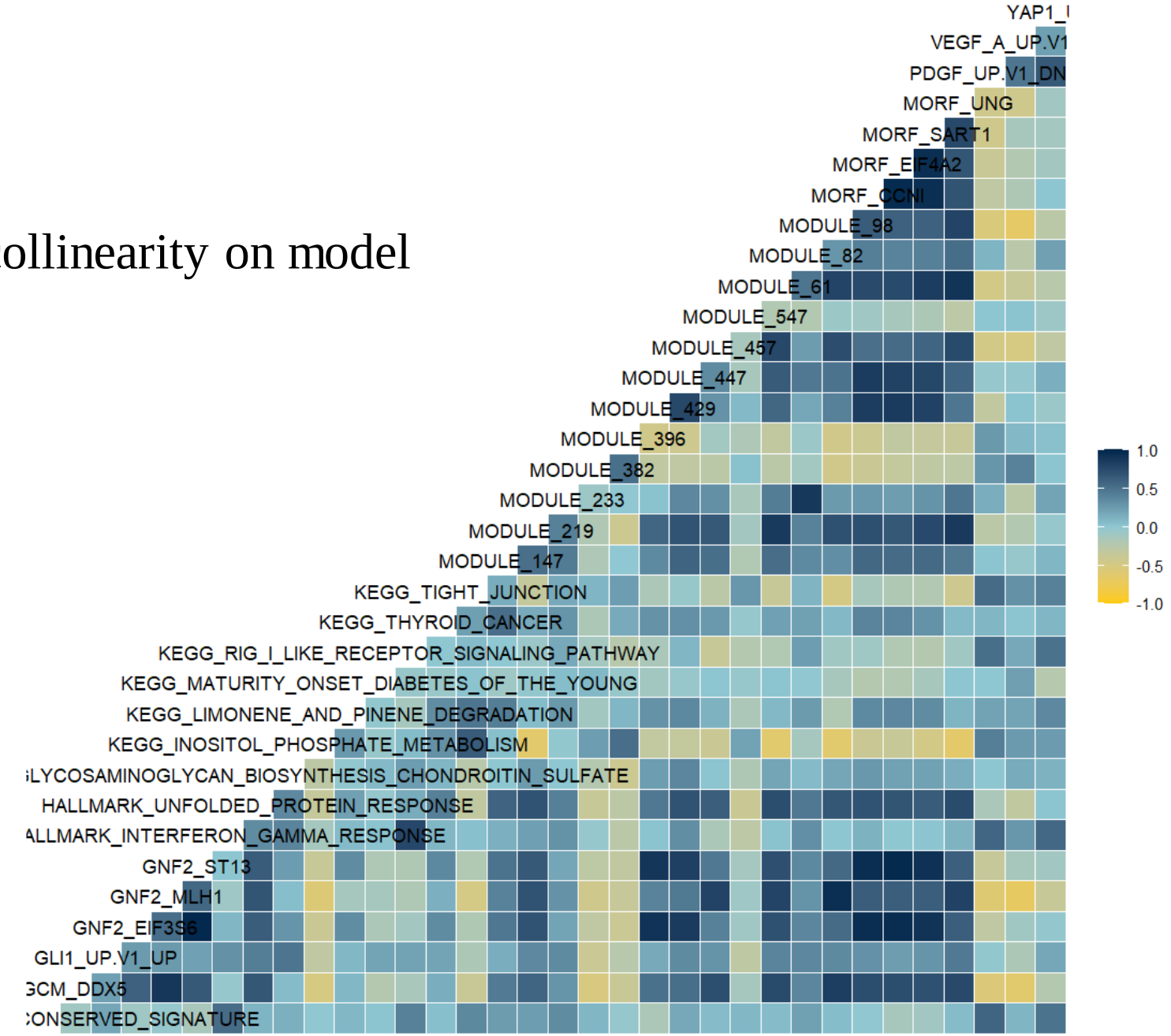


Ridge Regression

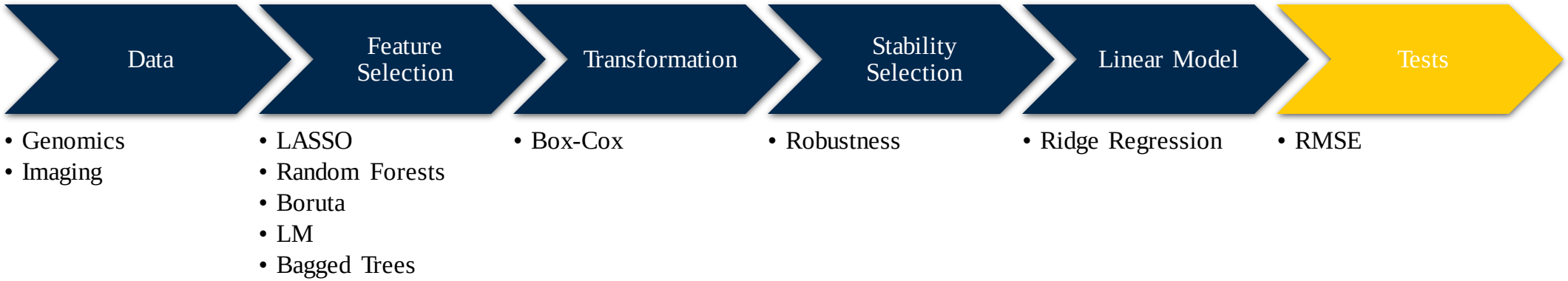
- Ridge Regression is a regularization technique used in linear regression models
- $\widehat{\beta_R} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top y$
- $\min_{\beta} (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \left|\left|\boldsymbol{\beta}\right|\right|_2^2$
- Regularization parameter (λ) controls penalty applied to coefficients
- Cross-validation finds optimal λ for best model performance

Why Ridge?

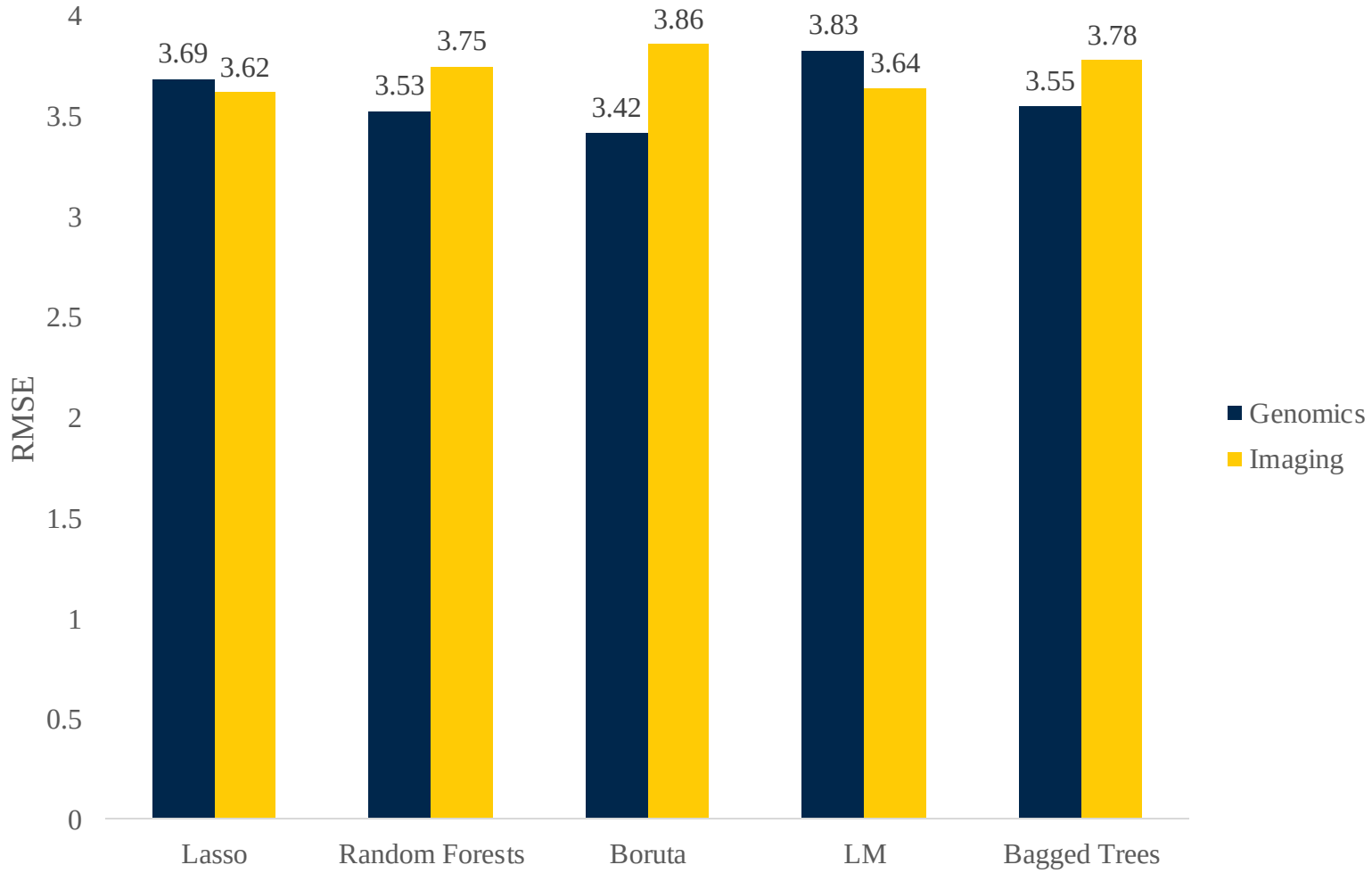
- Reduces impact of multicollinearity on model
- Prevents over-fitting



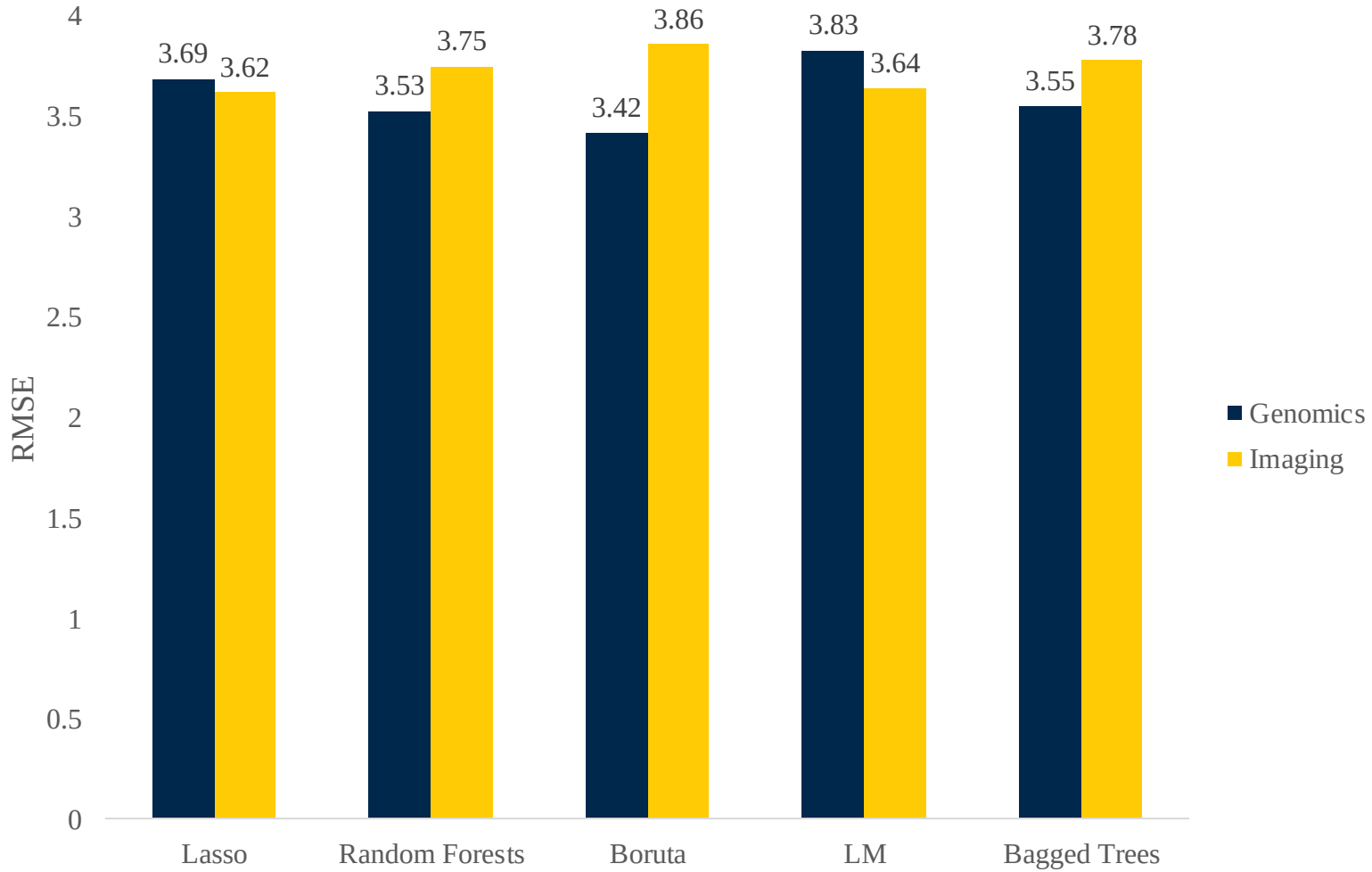
Pipeline of Project



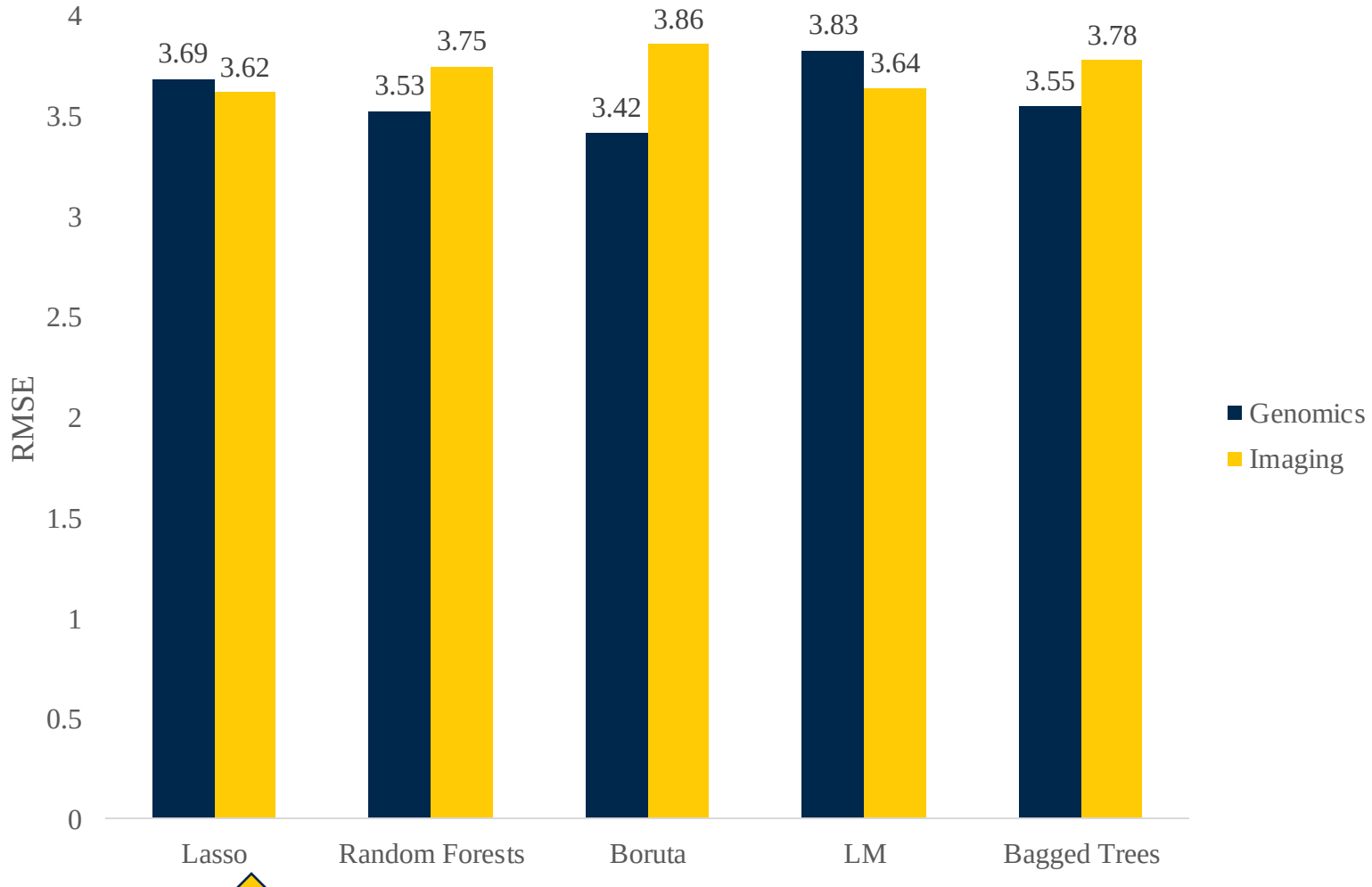
RMSE of Feature Selection Methods



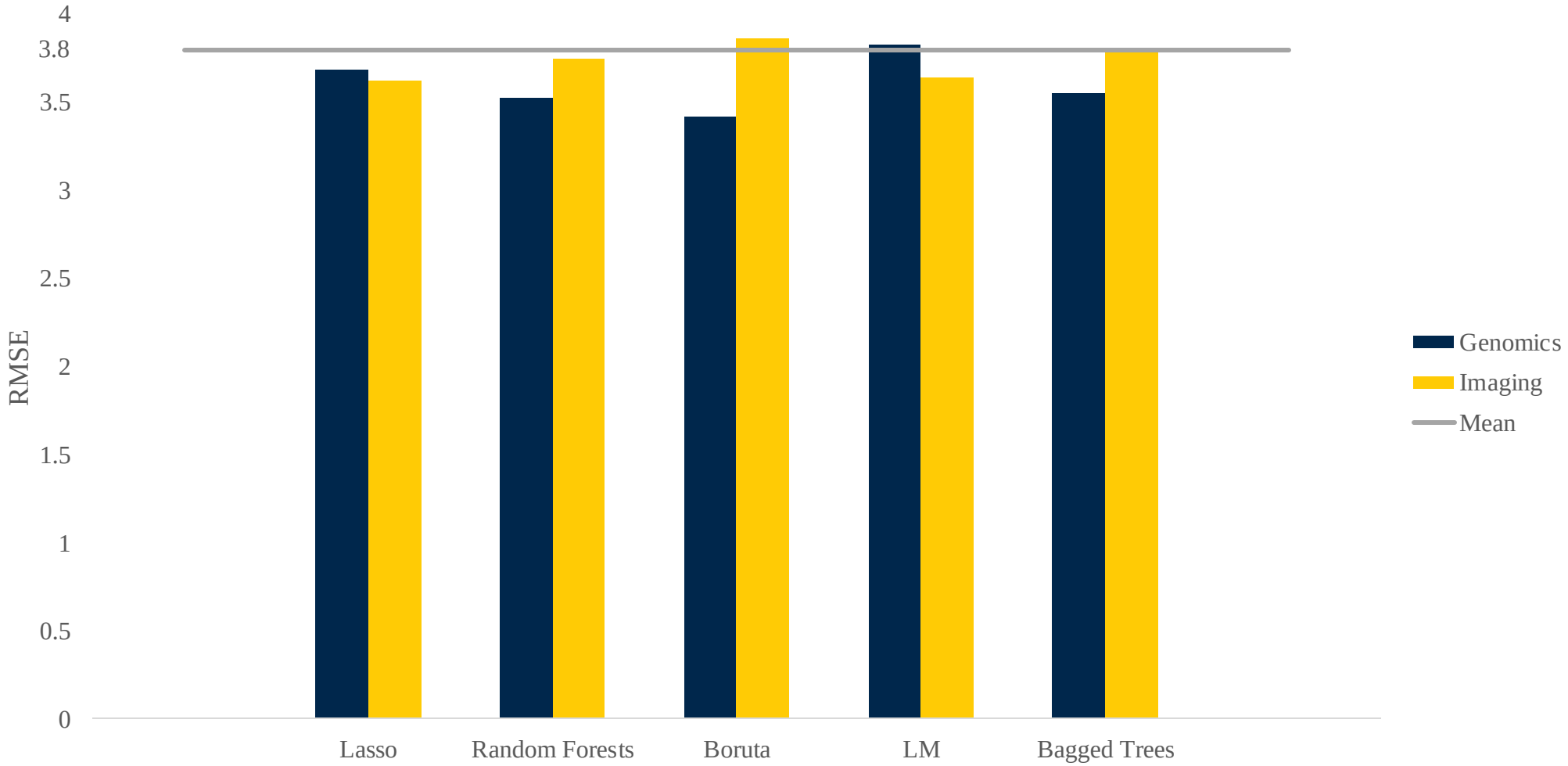
RMSE of Feature Selection Methods



RMSE of Feature Selection Methods

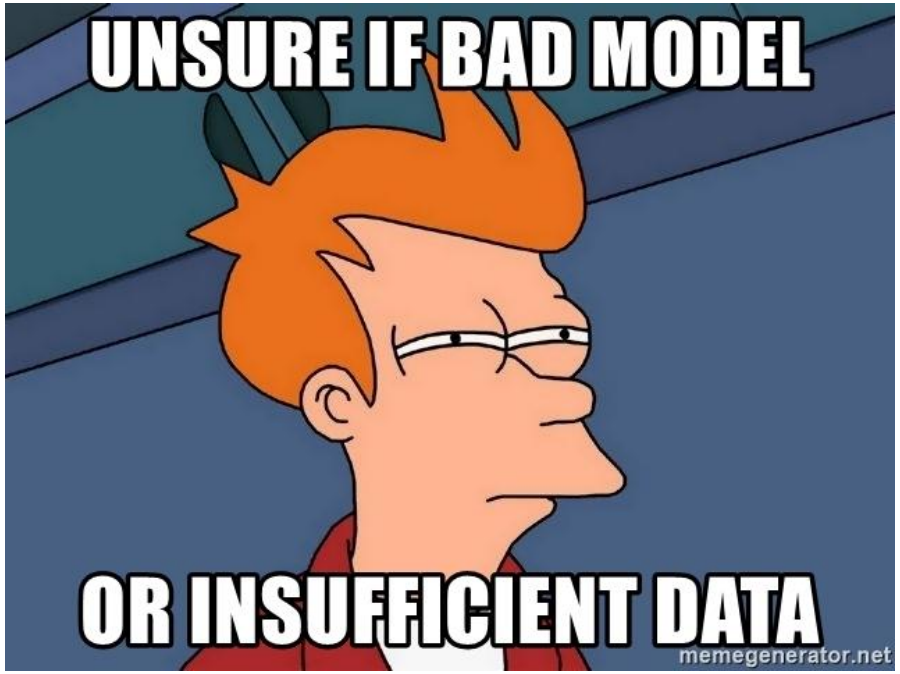


RMSE of Feature Selection Methods



Limitations

- Feature selection methods only slightly out-performed mean
- Small sample size
  - Seed dependence
- High dimensional data
  - $p \gg n$
- Unable to address multicollinear before feature selection
- Computationally expensive methods
- Limited information on disease severity/progression in patients
- Short Timeline





Takeaways

- Boruta method performed best overall with genomic data, but was highly variable between datasets
  - RMSE of 3.42
- LASSO is very consistent
  - 0.13 RMSE difference between pre-and post-stability models
- Stability selection effectively ensures reliability and robustness
  - Reduces the impact of dependency on randomness, especially for low *n*
- High dimensional data is very difficult to work with

References

Breiman, L. (1996). Bagging predictors. Machine learning, 24, 123-140.

Breiman, L. (2001). Random forests. Machine learning, 45, 5-32.

Huang, S., & Marchetti-Bowick, M. (2014). Summary and discussion of: "Stability Selection" Statistics Journal Club, 36-825.

Liaw, A., & Wiener, M. (2002). Classification and regression by random forest. R news, 2(3), 18-22.

Meinshausen, N., & Bühlmann, P. (2010). Stability selection. Journal of the Royal Statistical Society Series B: Statistical Methodology, 72(4), 417-473.

Kursa, M. B., & Rudnicki, W. R. (2010). Feature Selection with the Boruta Package. Journal of Statistical Software, 36(11), 1–13. https://doi.org/10.18637/jss.v036.i11